

# Nick Romano

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## SUMMARY

Interactive Media Designer and Researcher merging engineering and the humanities to build intuitive interfaces, data systems, and collaborative workflows. My professional trajectory spans roles as an Industrial Automation Hardware Test Engineer and Product Owner and ASU Information Management Specialist, where I spearheaded the development of relational databases, data tools, and REST API communications to automate systems, streamline institutional reporting, and visualize complex student lifecycle metrics for high-level stakeholders while completing my undergraduate studies. Early adopter of LLM agents and AI development workflows (Claude), utilizing these tools to rapidly prototype software utilities.

A Master's in Media Studies and a background in K-5 and university STEAM education inform my interdisciplinary approach to design, interweaving digital ethnography, media archaeology, platform studies, low-fi prototyping, software development, and iterative design. Additionally, a decade of jazz performance shapes my highly collaborative methodology toward group research and design thinking. Recent presenter and exhibitor at IEEE/CVF CVPR 2026 and UPenn's Rotunda.

## EDUCATION

**Parsons School of Design • The New School • New York, NY**

**Master of Arts in Media Studies • Aug 2024 – May 2026**

- **Cumulative GPA:** 4.0/4.0
- **Thesis Abstract:** Techno-juggling
- **Awards:** MA, Media Studies Distinguished Thesis Award, May 2026.

**Advanced Computing Center for the Arts and Design (ACCAD), The Ohio State University • Columbus, OH • Jan 2024 – May 2024**

- **Graduate-Level Coursework:** Designing Experiential Media Systems
- **GPA:** 4.0/4.0

**Ira A. Fulton Schools of Engineering • Arizona State University (ASU) • Tempe, AZ**

**BS in Electrical Engineering • Aug 2018 – Dec 2022**

- **Cumulative GPA:** 3.9/4.0
- **Capstone:** A Full-Stack Interface for Residential Solar Energy Viability
- **Awards:** Summa Cum Laude

## SKILLS

- **Web Development:** HTML5, CSS3, JavaScript, Responsive Design, Web & Mobile Applications
- **Programming:** Python, API Integration, Source Control (Git), Claude.AI, Electron, NodeJS, A-Frame
- **Data:** Data Visualization (Plotly & Three.js), Data Analysis, REST APIs, AstraDB, Excel
- **Other:** Interactive Systems, Soldering, Arduino, HW Testing, Augmented Reality, Projection Mapping, Computer Vision Object Tracking, ProTools, Adobe Premiere, Unity, Blender, MIDI, Touch Designer, Jira

## TEACHING AND RESEARCH

**Fellow and Technical Lead, Techno-juggling (Interactive Installation) • Ethnographic Media Lab, The New School for Social Research • Cleveland, OH • Oct 2025 – May 2026**

- **Showcased Techno-juggling** to a diverse audience of peers and faculty at The New School's Work-in-Process exhibition, New York, NY, April 2026. Educating participants on the mechanics of computer vision and real-time data mapping.
- **Architected a cross-platform interactive environment** using **Electron** and **WebSocket**, enabling real-time spatial data visualization for public educational displays.
- **Managed iterative design cycles** based on documented fieldwork and user testing, translating complex technical feedback into accessible interactive features.
- **Leveraged media archaeology methods** to inform system design, exploring the historical intersections of performance art and emerging technology.

**Coding Tutor • The New School • Remote • Jan 2026 – May 2026**

- **Facilitated advanced technical practicums for Media Studies students**, translating core programming concepts (HTML5, CSS3, JavaScript) into creative, project-based applications.
- **Mentored student-led prototyping**, providing high-level debugging and architectural guidance for interdisciplinary media installations and web-based art.
- **Developed a scalable e-learning curriculum**, authoring a library of instructional documentation and video tutorials to standardize technical literacy for future academic cohorts.

**Projection Mapping Designer • Jason Toth's Creative Studio • 78th Street Studios, Cleveland • Jan - March 2026**

- **Developed Touchdesigner Programs** to shape multi wall-projection mapped experiences of artist's work and animations for "Listening Room" guests.
- **Followed Iterative Design Processes** based on artist's feedback moving from external rectangular window based mappings to a set of seven triangular arcs with audiovisual feedback on the wooden frames.
- **Led Technical Problem Solving** to mount a short-throw projector into the room and implemented a phone based player connection with Touchdesigner such that artist could queue up YouTube videos in situ.

**Project Lead and Grant Recipient • Renaissance Box • Cleveland, OH • Jun 2025 – Sep 2025**

- **Project Leadership and Funding:** Secured a \$1,000 grant through the Renaissance Box incubator to spearhead the development of large-scale interactive media installations for the **Cleveland Ingenuity Festival** and the **Cleveland Institute of Art's Interactive Media Lab**.
- **Team Management:** Directed a multidisciplinary team including a mixed-media artist and a high school mentee, overseeing the project lifecycle from conceptualization to public exhibition.
- **Iterative Design:** Employed a rigorous R&D workflow featuring **concept mapping, rapid prototyping, and field research**, utilizing community playtesting to refine user UX/UI and hardware reliability.

### **Key Installations:**

- **Collage-Bop:** Developed a computer-vision-based installation where participants manipulate air hockey pucks to generate real-time video collages.
  - Engineered a system using an **overhead camera to track puck coordinates**, mapping spatial data to a user-facing visual interface.
  - Integrated **MIDI keyboard controls** allowing users to modulate audio-visual parameters including volume, geometry, layering, and playback speed.
  - Developed a post-interaction pipeline where users received a unique code to replay their specific creation process via a dedicated web portal.
- **Tell-A-Vision:** Designed and fabricated a robotic human figure inspired by Nam June Paik, constructed from upcycled CRT monitors, telephones, and mobile devices.
  - Programmed limb-mounted monitors to display **dynamic visual content triggered by participant motion**, blending analog aesthetics with modern sensor-based interactivity.

**AR Developer, Broadway Theatre's 1920s Marquee Revival • Mount Pleasant, MI • May - Aug 2025**

- **Augmented Reality Development:** Engineered a mobile-based WebAR experience designed to drive community engagement and fundraising efforts for the restoration of the historic marquee.
- **3D Modeling and Archival Research:** Conducted extensive historical research using the theatre's physical archives to accurately reconstruct digital assets; utilized **Blender** to model period-accurate 3D marquee components based on original source images and descriptions.
- **Geospatial Implementation:** Traveled on-site to perform **3D Visual Positioning System (VPS) scans** of the theatre facade, ensuring precise digital-to-physical alignment for the AR overlay.
- **WebAR Deployment:** Developed and optimized the mobile application using **Niantic 8th Wall**, enabling a seamless, browser-based user experience without the need for external app downloads.

**AR Developer, Remix'n' • Cleveland Museum of Art Transformer Station • Cleveland, OH • May - Aug 2025**

- **Interactive Exhibition Design:** Developed and deployed a custom Augmented Reality (AR) experience exhibited at the **Cleveland Museum of Art's Transformer Station**, bridging physical print media with digital interactivity.
- **WebAR Engineering:** Architected a browser-based application utilizing **Mind.AR, Three.js, and A-Frame** to deliver high-performance camera tracking and real-time 3D overlays without requiring a native app install.
- **AI-Augmented Development:** Leveraged **Claude.AI** as a technical collaborator to accelerate the coding process, optimizing JavaScript frameworks for seamless audio-visual synchronization and interactive link integration.
- **User Experience (UX):** Streamlined audience entry-points via QR-code activation, enabling users to instantly trigger immersive overlays that transform static prints into dynamic, multi-sensory experiences.

**Interactive Design Educator Science, Technology, Engineering, Arts, and Mathematics (STEAM) • Cleveland Play House • Cleveland, OH • May - July 2024**

- **Product Engineering & Fabrication:** Designed and constructed "Music Circle," a tactile "musical carpet" interface using vinyl records and conductive hardware integrated with Makey-Makey microcontrollers to facilitate simultaneous play for 8+ students.
- **Software Development:** Engineered a Python-based software suite to drive audio-visual educational games, implementing custom response logic to promote collaborative social learning and digital literacy.
- **Curriculum Leadership:** Developed a research-based STEAM framework and web-enabled instructional environments to deliver immersive tech curricula to K–4 students.
- **Instructional Management:** Led 25-student cohorts through a rigorous 5-week accelerated summer enrichment program, bridging the gap between complex hardware and early-childhood accessibility.
- **Iterative R&D:** Optimized installation engagement and durability through a rapid prototyping lifecycle, utilizing direct student feedback loops and situational design processes.

**Interactive Digital Systems Developer • OSU ACCAD • Columbus, OH • Jan – May 2024**

- **Researched and engineered interactive environments** using **Isadora**, specializing in real-time projection mapping and sensor-based media triggers.
- **Developed a "Snare-Drum Video Controller"** by interfacing analog percussion with digital outputs via **Makey-Makey**, demonstrating expertise in hardware-software integration.
- **Prototyped a generative audiovisual "Poetry Machine"** and a time-synced Bebop historical narrative, utilizing complex logic to synchronize multi-channel media.
- **Applied iterative design methodologies** under the mentorship of Alex Oliszewski, refining UI/UX principles for immersive, non-traditional performance spaces.

**Lead Web Systems Developer, Engineering Capstone • ASU • Tempe, AZ • Jan – Dec 2022**

- **Architected a full-stack interactive dashboard** to quantify solar panel viability, translating complex geospatial data into actionable consumer insights.
- **Engineered a multi-source API pipeline** in Python, synthesizing real-time data from Bing Maps, Google Maps, and the NREL (National Renewable Energy Laboratory) database.
- **Developed a high-performance UI** using the Dash (Plotly) framework and custom CSS, ensuring a responsive experience for complex data visualization.

- **Designed a robust relational database schema** to manage and query location-based solar metrics, optimizing data retrieval speeds.
- **Directed the technical lifecycle**, from initial competitive research and UI/UX analysis to final deployment and stakeholder presentation.

**Physics I Instructional Assistant • Ira A. Fulton Schools of Engineering • ASU • Tempe, AZ • Aug - Dec 2019**

- **Optimized student learning outcomes** for Physics I curriculum through the strategic design of supplemental lesson plans and comprehensive exam prep modules.
- **Facilitated high-impact tutoring sessions** for small cohorts (1–5 students), utilizing targeted intervention strategies to clarify Newtonian mechanics and thermodynamic principles.
- **Synthesized complex theoretical concepts** into digestible instructional assets, including curated lecture notes and visualized problem-solving guides.

## **ENGAGEMENTS**

- **Exhibitor & Speaker • IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)**, a premier annual computer vision event • Denver, CO • Jun 2026.
  - Showcased Techno-juggling for 3 days in the conference's AI Art Gallery.
  - Selected from 260 submissions; presented at a 4,000+ attendee industry and academic event.
  - Spoke on a panel regarding the role of AI in design and creative practice.
- **Featured Performer & Speaker • Technical Salon and Algorave**, a math-driven generative composition • The Rotunda (UPenn) • Philadelphia, PA • May 2026.
  - Synthesized complex code and aesthetic theory into an interactive "salon" format to encourage peer-to-peer technical learning. Showcased technical mastery in mixing diverse media modalities through live algorithmic performance.

## **PROFESSIONAL EXPERIENCE**

**Gallery Attendant • Cleveland Museum of Art (CMA) • Cleveland, OH • Oct 2024 – Oct 2025**

- **Acted as a technical and cultural translator**, guiding a diverse public through the intersection of traditional art collections and new-media digital exhibitions.
- **Monitored and optimized visitor interaction** within high-traffic zones, maintaining a safe environment while fostering deep engagement with interactive media.
- **Selected by the Cleveland Museum of Art's Exhibition Department for the *CMA Artists at Work* showcase**; developed and deployed a custom Augmented Reality (AR) experience.

**Electrical Engineering Product Owner • Rockwell Automation • Cleveland, OH • Jun 2022 – Jan 2024**

- **Promoted to Product Owner** within three months of completing co-op.
- **Led the integration of complex web services**, bridging Jira, Jenkins, and Jama to build a robust analytics infrastructure for organizational reporting.
- **Developed custom automation frameworks** using Python APIs, significantly increasing the efficiency of technical testing and data collection processes.
- **Facilitated cross-functional collaboration**, mentoring and directing a diverse team of 10 engineers in an Agile environment.
- **Designed interactive data visualization dashboards**, translating internal engineering metrics into accessible insights for broad institutional visibility.

**Information Management Specialist • Arizona State University • Tempe, AZ • Aug 2019 – Jan 2022**

- **Spearheaded the development of data tools** focused on tracking student lifecycle engagement and enhancing program-level education initiatives.
- **Automated departmental communications** utilizing REST APIs, increasing the efficiency of digital content distribution to students and faculty.
- **Leveraged data visualization** to communicate program impact and student outcomes during high-level stakeholder meetings.
- **Managed relational database systems** to streamline the collection and analysis of student metrics, directly supporting institutional reporting requirements.